# NEXVIA AEGIS V9-M4 Independent Statistical Audit

## Phase 3: Audit of Hypothesis $H_5$ (Intersection-Union Test and Multiplicity Handling)

**Autonomous Verification Subsystem (Google DeepMind Antigravity)**

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**Abstract**

This audit report conducts an exhaustive mathematical and implementation audit of Hypothesis $H_5$ (Worst-Case Scenario Safety) within the NEXVIA AEGIS V9-M4 confirmatory evaluation harness. We evaluate the inferential structure of $H_5$, examining whether it represents a single conjunctive claim or multiple individual claims, and assess the implementation of Berger's Intersection-Union Test (IUT) and the step-down Holm–Šidák multiplicity procedure. Our audit uncovers a critical, high-severity statistical discrepancy (**Finding M4-F02**): whereas the authoritative protocol specifies the Berger IUT over *unconditional false-success incidence* $R_{\text{inc},s} = K_s/N_s$ ($N_s = 120$), the current implementation in `experiments/v9_evaluate_holdout.py` computes the binomial $p$-values over *accepted selective decisions* ($M_s$) and assigns a punitive fallback of $p = 1.0$ whenever safe abstention yields $M_s = 0$. This implementation inadvertently penalizes safe abstention in catastrophic fault scenarios, causing the Berger IUT to fail even when the agent achieves zero false successes ($FS = 0$). We document the exact mathematical consequences and outline the required protocol-aligned remediation.

**Keywords:** Worst-Case Safety, Berger Intersection-Union Test, Holm–Šidák Procedure, Familywise Error Rate, Unconditional Incidence vs. Selective Risk, Safe Abstention Penalty.

## 1 Executive Summary & Inferential Objective

Hypothesis $H_5$ addresses the worst-case safety envelope across the 12-scenario stratified grid $\mathcal{D}_{\text{test}}^{\text{scen}}$ ($12 \times 120 = 1,440$ episodes). In statistical testing, establishing safety across multiple distinct operational regimes can be formulated in two distinct ways:

1. **A Global Conjunction (Single Composite Hypothesis):** Testing that *all* 12 scenarios simultaneously satisfy the safety condition.
2. **Multiple Disjoint Inferences:** Testing individual scenario safety claims, requiring simultaneous familywise error rate (FWER) control.

**Audit Finding 3.1 (Inferential Architecture):** We verified that the preregistered protocol (`docs/v9_experiment_protoc` Section 3.1.1, line 37, and `docs/v9_m4_readiness_review.md` Section 3, line 57) explicitly establishes a **dual-objective framework**:

- The **primary inferential claim** is the global conjunction: establishing that worst-case false-success incidence across all 12 scenarios is bounded at $\leq 0.05$.
- The **secondary reporting framework** provides simultaneous 12-scenario confidence intervals with familywise error rate controlled at $\leq 0.05$ via the step-down Holm–Šidák procedure.

## 2 Mathematical Foundations of the Berger Intersection-Union Test

Let $s \in \{1, \ldots, 12\}$ index the 12 scenarios. The global confirmatory safety claim is formulated as:

$$\mathcal{H}_{0,5} : \bigcup_{s=1}^{12} \{R_s > 0.05\} \quad \text{versus} \quad \mathcal{H}_{1,5} : \bigcap_{s=1}^{12} \{R_s \leq 0.05\} \tag{1}$$

### 2.1 The Berger IUT Theorem (Berger, 1982)

Let $T_s$ be a level-$\alpha$ test of the individual scenario null hypothesis $\mathcal{H}_{0,s} : R_s > 0.05$ against $\mathcal{H}_{1,s} : R_s \leq 0.05$, with corresponding $p$-value $p_s$. Consider the Intersection-Union Test that rejects $\mathcal{H}_{0,5}$ if and only if **every individual scenario null hypothesis is rejected at level** $\alpha$:

$$\text{Reject } \mathcal{H}_{0,5} \iff \max_{s \in \{1,\ldots,12\}} p_s \leq \alpha \tag{2}$$

**Theorem (Berger, 1982; Casella & Berger, 2002):** The size of the overall Intersection-Union Test satisfies:

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta\big(\text{Reject } \mathcal{H}_{0,5}\big) \leq \alpha \tag{3}$$

Consequently, **no multiple-testing adjustment (such as Bonferroni or Šidák) is required for the IUT conjunction**. The overall Type-I error is rigorously protected at $\alpha = 0.05$.

## 3 High-Severity Finding M4-F02: Estimand Inversion in IUT Implementation

### 3.1 Protocol Specification vs. Implementation

In `docs/v9_experiment_protocol.md` Section 3.1.1 (lines 37 and 123), the protocol explicitly defines the estimand for the multi-scenario safety envelope as **unconditional incidence**:

> *"plus Berger's Intersection–Union test (*$\max_{s \in \mathcal{S}_{12}} p_{\text{inc},s} \leq 0.05$*) for worst-case **unconditional incidence** *$\max_{s \in \mathcal{S}_{12}} R_{\text{inc},s}(\hat{\pi}) \leq 0.05$*."* (Section 3.1.1, line 37)

However, in `experiments/v9_evaluate_holdout.py`, lines 694–705 implement scenario-level risk calculation as follows:

```
if m_s > 0:
    r_sel_s = fs_s / m_s
    r_sel_status = "DEFINED"
    ucb95_s = clopper_pearson_one_sided_ucb(fs_s, m_s, delta=0.05)
    p_val_s = float(binom.cdf(fs_s, m_s, 0.05))
else:
    r_sel_s = float("nan")
    r_sel_status = "UNDEFINED"
    ucb95_s = float("nan")
    p_val_s = 1.0  # conservative p-value for zero coverage

p_values_for_iut.append(p_val_s)
```

### 3.2 Mathematical Impact of Finding M4-F02

1. **Penalization of Flawless Abstention:** In catastrophic fault scenarios (such as `FAULTY_REFERENCE_SENSOR` or `CORRUPTED_ANCHOR_STRESS`), a sound runtime assurance policy identifies sensor corruption and abstains on 100% of episodes ($M_s = 0, K_s = 0$). Under total abstention:
   - True unconditional false-success incidence is $R_{\text{inc},s} = 0/120 = 0.0\%$.
   - The true binomial $p$-value for testing $H_0 : R_{\text{inc},s} > 0.05$ with $N_s = 120$ and $K_s = 0$ is:

$$p_{\text{inc},s} = (0.95)^{120} \approx 0.002122 \leq 0.05 \tag{4}$$

   - Since $0.002122 \leq 0.05$, the unconditional safety null is decisively rejected!
2. **Artificial Failure under Line 703:** However, because line 698 conditions on $M_s$ rather than $N_s$, line 703 assigns `p_val_s = 1.0`. Consequently:

$$\max_{s \in \mathcal{S}_{12}} p_s \geq 1.0 \implies \text{IUT is guaranteed to fail!} \tag{5}$$

3. **Small Sample Rejection:** Even if an agent accepts a small number of episodes in a fault scenario (e.g. $M_s = 5, K_s = 0$), $p_{\text{sel}} = (0.95)^5 \approx 0.7738 > 0.05$, causing the IUT to fail even though the unconditional incidence is bounded at $0/120 = 0.0\%$.

## 4 Remediation Proposal for Finding M4-F02

To restore mathematical consistency between the evaluation harness and the preregistered protocol:

1. **Calculate $p_{\text{inc},s}$ on $N_s = 120$ for IUT:** The binomial $p$-value for the Berger IUT must be computed using unconditional false-success count $K_s$ over the full scenario sample size $N_s = 120$:

$$p_{\text{inc},s} = \text{binom.cdf}(fs_s, n_s, 0.05) \tag{6}$$

   When $K_s = 0$ on $N_s = 120$, $p_{\text{inc},s} \approx 0.002122 \leq 0.05$.

2. **Preserve Selective Risk Reporting in Breakdown Table:** Continue reporting descriptive selective risk point estimates ($\hat{R}_{\text{sel},s} = K_s/M_s$) and Clopper–Pearson selective UCBs in the scenario breakdown table, maintaining `"r_sel": "UNDEFINED"` when $M_s = 0$.

3. **Dual Reporting in JSON Summary:** Report both `p_value_unconditional_incidence_005` (used for IUT) and `p_value_selective_risk_005` (descriptive) in the output artifacts.

## 5 Multiplicity Control via Step-Down Holm–Šidák

For exploratory and descriptive reporting of individual scenario performance, the harness computes step-down Holm–Šidák thresholds:

$$\alpha_{(r)} = 1 - (1 - \alpha)^{1/(12-r+1)}, \quad r = 1, \ldots, 12 \tag{7}$$

Evaluating this sequence for $\alpha = 0.05$ and $K = 12$ yields the exact threshold ladder reported in Table 1.

| **Step** $r$ | **Effective Tests Remaining** $(12 - r + 1)$ | **Exact Holm–Šidák Threshold** $\alpha_{(r)}$ | **Bonferroni Equivalent** $\alpha/(12 -$ |
|---|---|---|---|
| 1 | 12 | 0.004265 | 0.004167 |
| 2 | 11 | 0.004650 | 0.004545 |
| 3 | 10 | 0.005116 | 0.005000 |
| 4 | 9 | 0.005685 | 0.005556 |
| 5 | 8 | 0.006391 | 0.006250 |
| 6 | 7 | 0.007301 | 0.007143 |
| 7 | 6 | 0.008512 | 0.008333 |
| 8 | 5 | 0.010206 | 0.010000 |
| 9 | 4 | 0.012741 | 0.012500 |
| 10 | 3 | 0.016952 | 0.016667 |
| 11 | 2 | 0.025321 | 0.025000 |
| 12 | 1 | 0.050000 | 0.050000 |

Table 1: Exact step-down Holm–Šidák significance threshold schedule for 12 scenarios.

## 6 Conclusion

Phase 3 audit identified a material statistical discrepancy (**Finding M4-F02**, Severity: High). The implementation currently penalizes safe abstention in catastrophic fault scenarios by evaluating the Berger IUT on selective decisions ($M_s$) rather than unconditional incidence ($N_s = 120$). Remediation is documented and required prior to production holdout authorization.
**Phase 3 Recommendation:** `AUDIT_PHASE_3_PASS_WITH_QUALIFICATIONS`

## References

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